A finite consistency check of Written-on-the-Wall conjectures 20 and 21 on all connected graphs with $4 \le n \le 11$

(the statement is claimed proved in the literature; not a new result)

Xavier Callens (SocrateAI Lab)

Preprint v1, 2026-11-01

doi:10.5281/zenodo.23856042 (not peer reviewed)

Zenodo DOI to be assigned on release (not peer reviewed)

AI-assistance disclosure and audit status. This preprint, its code, its claim ledger and the audit were produced by AI agents (Claude, Anthropic) under the author's direction; the author is responsible for the content. **This version was audited by a separate AI agent instance (`AUDIT.md`, ACCEPT_WITH_CORRECTIONS), not by a human.** The audit's eight corrections are applied (Section 6). No human audit exists. There is no Lean in this paper.

Abstract

Frame. Since the adjacency matrix has trace 0, $S^+ = E/2$, where S^+ is the sum of the positive eigenvalues and E the graph energy. Written-on-the-Wall (WoW, Graffiti) conjectures 20 ($n_+ \le S^+$) and 21 ($n_- \le S^+$) are therefore, together, exactly $E(G) \ge 2\max(n_+, n_-)$. Kumar and Pragada (arXiv:2607.19817, 22 Jul 2026, *unrefereed*, AI used for ideation; we did *not* check the proof) claim $E(G) \ge 2(n - \alpha(G))$ (Fajtlowicz's Graffiti #543) and, with the classical inertia bound, deduce $E \ge 2\max(n_+, n_-)$; they state that this resolves the Aouchiche–Hansen 2010 conjectures #20, #21. The smaller-count version ($S^+ \ge \min(n_+, n_-)$) has been a theorem since Fajtlowicz 1987 ($E \ge$ rank). **This census is a finite consistency check of a claimed theorem, not a new result.**

What was run. All 1,018,690,325 connected graphs with $4 \le n \le 11$ (nauty `geng -c`) were scanned in double precision (Tier [X]): 0 violations of 20 or 21. Conjecture 21 is tight exactly at K_n among connected graphs ($n \le 10$ exhaustively, $n = 11$ via a flag band); each tie was decided exactly (Tier [C]). The smallest *non-*K_n margins for 21 are certified and shrink with n: $0.2143, 0.1463, 0.1249, 0.0941, 0.0806, 0.0645, 0.0561$ for $n = 5..11$. An independent audit recomputed all $n \le 10$ and 12.2% of $n = 11$ with different code and exact inertia, matching with 0 violations.

What this is NOT. Not a new result and not a proof; not a statement about $n \ge 12$; not interval-certified (the scan uses floating point, Tier [X]; soundness rests on an argument, Section 4); not formalised in Lean. The producer's earlier note "nothing found for 20, 21" was a literature miss.

Markers. [L]: a cited source, taken as stated. [C]: executed exact code with the argument written down, not machine-checked. [X]: floating-point or recorded-absence evidence. Identifiers `GR11-·-nnnn` refer to `ledger/ledger.json`.

1 Statement, frame and priority

For a graph G with adjacency eigenvalues λ_i let n_+, n_- be the numbers of strictly positive and negative eigenvalues (with multiplicity), and $S^+ = \sum_{\lambda_i > 0} \lambda_i$. WoW (July 2004, p. 19; read by eye from a scanned bitmap) states: **20**: $n_+ \le S^+$; **21**: $n_- \le S^+$. Since $\operatorname{tr} A = 0$, $E = \sum |\lambda_i| = 2S^+$, so 20 and 21 together are $E \ge 2\max(n_+, n_-)$.

- **Claimed proof ([L]).** Kumar–Pragada [1], Thm. 1.2: $E(G) \geq 2(n-\alpha(G))$. With $\alpha \leq \min(n-n_+, n-n_-)$ this gives $E \geq 2\max(n_+, n_-)$, which they say "resolves [Aouchiche–Hansen, Conjectures #20, #21, Table 6]". The deduction is elementary; Theorem 1.2 is an unrefereed preprint whose proof we did not check.
- **Older partial result ([L]).** The WoW comment on p. 19 cites Fajtlowicz 1987: $E \geq \text{rank} = n_+ + n_-$, i.e. $S^+ \geq \min(n_+, n_-)$ for every graph. Hence whichever of 20, 21 has the smaller count has been a theorem since 1987.
- **Not evidence of openness.** Zhang [2] calls "$E \geq 2\min(n_+, n_-)$" open; this is evidently a slip for max and should not be read as evidence that the problem is open. Roucairol–Cazenave [3] do not treat 20 or 21.

The statements are for all graphs; the scan covers connected graphs only, which loses nothing since n_+, n_-, S^+ are additive over components.

2 Method

Scan ([X]). `data/code/scan.py`: `nauty-geng -c -q n [res/mod]`, batched graph6 decoding, `numpy.linalg.eigvalsh`; $n_\pm$ count eigenvalues beyond $\pm 10^{-9}$, S^+ the sum of those above 10^{-9}. Margins $m_{20} = S^+ - n_+$, $m_{21} = S^+ - n_-$. A violation is $m < -10^{-6}$; graphs with $m < 10^{-6}$ are written to a flag file. For $n = 11$ the 1,006,700,565 graphs were split into 40 `geng` residue classes (cost 12.48 core-hours, cores 4, 5 only); the part counts sum to exactly the OEIS A001349 value. Graph counts for $n = 4..11$: $6, 21, 112, 853, 11117, 261080, 11716571, 1006700565$.

Exact decision ([C]). `exact_check.py`: integer characteristic polynomial (sympy), exact inertia by Descartes' rule (valid because the polynomial is real-rooted), S^+ enclosed by rigorous root isolation. *Deviation:* the addendum promised Sturm sequences; the code uses Descartes' rule, equivalent for real-rooted polynomials (audit correction 8).

Controls. Positive: Graffiti 29 (WoW p. 23: Randić index $\leq$ number of negative *distance* eigenvalues, "disproved by Alon, Saks, Seymour, Shearer, Winkler") must first fire at $n = 7$ and not for $n \leq 6$: it does (graph6 `FEhf?`; counts $0, 0, 0, 1, 1, 23, 18$ for $n = 4..10$, 1862 at $n = 11$). It caught two decoding bugs in the first version of the code, fixed before any reported number. Negative: $\lambda_1 \geq$ average degree and $\sum |\lambda| \geq n_+ + n_-$ are theorems and fired 0 times.

3 Results

n	connected graphs	viol. 20 / 21	min m_{20}	min non-K_n m_{21} (certified)
4	6	0 / 0	0.2361 (P_4)	0.2361 (P_4)
5	21	0 / 0	0.2361	0.2143 `DQw`
6	112	0 / 0	0.4495	0.1463 `EQjO`
7	853	0 / 0	0.4587	0.1249 `FQhVO`
8	11117	0 / 0	0.5388	0.0941 `GQhTUg`
9	261080	0 / 0	0.6631	0.0806 `HQhTQji`
10	11716571	0 / 0	0.6538	0.0645 `IQhTQiitO`
11	1006700565	0 / 0	0.8283	0.0561 `JQhTQiiTVT?` (part 39)

Table 1: Scan results ([X]) and the certified margins ([C]; sympy enclosures of width 10^{-25}, re-run for this paper, `data/audit/out/exact_margin_rerun_2026-11-01.txt`). The producer's `min_gap21` statistic included K_n and could not show the shrinking margin. The $n = 11$ minimum for 21 over the other 39 parts is the producer's float value; part 31 (which contains K_{11}) has non-K_{11} minimum 0.465 (auditor).

Scope. For $n \le 3$: K_1 (all zero) and K_2 (spectrum $\{1, -1\}$) give equality in *both* 20 and 21, K_3 gives equality in 21, P_3 is strict. So the "smallest margin 0.236 for 20" holds for $n \ge 4$ only (attained at P_4); over all connected graphs 20 is tight at K_1, K_2. Hence the range $4 \le n \le 11$.

Equality. K_n has spectrum $\{n-1, (-1)^{n-1}\}$, so $S^+ = n - 1 = n_-$: equality in 21, exact by hand and confirmed by the integer characteristic polynomial for K_4, K_9, K_{10}, K_{11}. Among connected graphs, 21 is tight only at K_n ($n \le 10$ exhaustively; $n = 11$ by the flag band: the single flagged graph over all 40 parts is K_{11}, in part 31, decided exactly). Disconnected unions of cliques also give equality. The raw strict < 0 float tally showed K_9, K_{10}, K_{11} at $-9 \cdot 10^{-16}$ to $-1.8 \cdot 10^{-15}$ (float noise), each decided exactly as equality. At $n = 11$ the smallest non-K_{11} margin for 21 is 0.05606 (part 39), not "none flagged within 10^{-6}".

4 Rounding soundness (whole census)

For an integer symmetric matrix the product of the nonzero eigenvalues is $\pm$ the lowest nonzero characteristic-polynomial coefficient, an integer of absolute value ≥ 1. The other $k \le 10$ nonzero eigenvalues have $\sum \lambda^2 \le 2m \le 110$, so by AM–GM their product is at most $(110/k)^{k/2} \le 11^5$. Hence every nonzero $|\lambda| \ge 1/161051 \approx 6.2 \cdot 10^{-6}$ for $n \le 11$. With backward error about 10^{-14} the 10^{-9} threshold classifies every eigenvalue correctly, so $n_\pm$ are exact for all $1.0 \cdot 10^9$ graphs *assuming LAPACK backward stability.* Weyl's inequality bounds the error of S^+ by about 10^{-12}, far below the 10^{-6} flag band. Misclassifying a true zero would only lower a margin (false flag), never hide a violation. The scan is nevertheless **not interval-certified** (Tier [X]). Corroboration (auditor, 134.8M graphs): 0 mismatches between float and exact inertia; smallest nonzero $|\lambda| = 0.0038$; largest `dgeev` vs `dsyevd` difference in S^+ is $4.8 \cdot 10^{-14}$. Regular graphs tie exactly at Randić $R = n/2$ in control 29, which is why its $+10^{-9}$ tolerance is needed.

5 The independent audit

A separate AI agent instance (not a person; `AUDIT.md`, ACCEPT_WITH_CORRECTIONS) recomputed with its own graph6 decoder, exact inertia from the integer characteristic polynomial (Faddeev–LeVerrier, guarded below 2^{52}, plus Descartes), S^+ from `dgeev` cross-checked with `dsyevd`, BFS distances, and certified sympy enclosures for the minimisers (cores 4, 5, about 4.9 core-hours). Results: all 11,989,760 graphs with $n \le 10$ match with 0 violations; $n = 11$ parts $31, 39, 12, 5, 7$ of 40 (122,811,341 graphs, 12.2%) match the producer's logs in every count and margin (to 10^{-14}); the producer's unmodified `scan.py 11 39 40` rerun is identical except the time. The other 35 $n = 11$ parts rest on the producer's run plus the argument of Section 4; the 1862 firings of control 29 were reproduced on 5 of 40 parts (245 firings). Control-29 firings at $n = 7, 8, 9, 10$ and four at $n = 11$ are genuine under exact distance-matrix inertia. Hashes of the three auditor scripts are in `AUDIT.md` and match the shipped copies.

6 Audit corrections applied

The producer's `data/RESULT.md` and `data/result.json` are kept unmodified (pre-audit); this paper supersedes them. (1) The novelty statement is replaced by Section 1. (2) Certified non-K_n margins for 21 are in Table 1. (3) The $n = 11$ smallest non-K_n margin is 0.05606. (4) The range is $4 \le n \le 11$. (5) The equality statement is restricted to connected graphs and K_n. (6) The zero-eigenvalue bound $6.2 \cdot 10^{-6}$ is added (Section 4). (7) The stale "$n = 11$ NOT run" text of `RESULT.md` (below its UPDATE) and "Auditor must redo 2" are superseded; $n = 11$ was run (all 40 parts). (8) Descartes versus Sturm is recorded as a deviation (Section 2).

7 Limitations and what is not done

- Kumar–Pragada's proof was not checked; it is an unrefereed preprint. Nothing of ours depends on it except the framing.
- AI audit only; no human audit. Tier [X] for the scan (not interval-certified); [C] for exact decisions; [L] for literature; no Lean.

- 35 of the 40 $n = 11$ parts and most of the control-29 total were not recomputed independently.
- WoW conjectures 39, 40, 129, 143, 154, 195, 284, 698 remain **BLOCKED** (definitions not pinned from the scan). Third-party claims (unrefereed GitHub material) that others are "already resolved" were not checked. Only 20, 21 and control 29 were pinned.
- Deviations from the preregistration (`PREREG_ITEM.md` sha256 `5a02b566...c3d85`; addendum 2 for $n = 11$, sha256 `3fd041d3...b10b`): Descartes instead of Sturm; the WoW source is a scanned bitmap read by eye (no OCR). The scan PDF is not shipped.

8 Claim ledger and reproducibility

`ledger/build_ledger.py` files the claims with hashed evidence; the Elenchus gate (`ledger/run_gate.py`, verbatim output in `ledger/gate_output.txt`) checks bookkeeping only and licenses nothing. Reproduce: `scan.py n res mod flagfile`; `exact_check.py 11 flagfile`; `run11.py w`; `agg11.py`; auditor: `audit_scan.py`, `exact_margin.py`, `exact_c29.py` (`data/audit/code`; run scripts as `.sh.txt`). Inputs are copies of `results/proposal5/2026-10-27/GR-11/` and of the auditor's scratch directory; nothing was committed, pushed or uploaded.

References

[1] H. Kumar, S. Pragada, *Energy and independence number*, arXiv:2607.19817 (v1, 22 Jul 2026), unrefereed; Thm. 1.2 and the corollary on #20, #21. ([L])

[2] S. Zhang, *Extremal values for the square energies of graphs*, arXiv:2409.15504 (2024). ([L])

[3] M. Roucairol, T. Cazenave, *Refutation of spectral graph theory conjectures with search algorithms*, arXiv:2409.18626 (2024). ([L])

[4] S. Fajtlowicz, *Written on the Wall* (July 2004 version), pp. 19, 23; Fajtlowicz 1987 as cited there; Aouchiche–Hansen 2010 as cited by [1]. Not re-read in the originals. ([L])

[5] B. D. McKay, A. Piperno, Practical graph isomorphism II, J. Symbolic Comput. 60 (2014) 94–112 (`geng`).